ESTRADIOL

Add progestogen if uterus intact

VTE, breast cancer risk

CHEMICAL NAME Estra-1,3,5(10)-triene-3,17β-diol (17β-oestradiol; valerate or hemihydrate)
CLASS Natural oestrogen (hormone replacement)
TRADE NAMES Estrofem · Progynova · Estrogel / Oestrogel · Vagifem (vaginal)

THERAPEUTIC USES / INDICATIONS

- Menopausal hormone replacement therapy (vasomotor symptoms, prevention of osteoporosis).
- Premature ovarian insufficiency, hypogonadism.
- Vaginal atrophy (low-dose vaginal tablets/cream).
- Endometrial preparation in frozen embryo transfer; gender-affirming therapy.

ADMINISTRATION

- **PO** daily; **transdermal** patch (twice weekly), gel (daily) — lower VTE risk.
- **Vaginal** tablets/cream/ring.
- Women with a uterus must also take a progestogen (endometrial protection).

DOSAGES

- **Oral HRT:** 1–2 mg daily.
- **Patch:** 25–100 mcg/24 h applied twice weekly.
- **Gel:** 0.5–1.5 mg daily applied to lower trunk or thigh (Sandrena; site varies by product).
- **Vaginal tablets:** 10 mcg daily for 2 weeks, then twice weekly.

PHARMACOKINETICS

- Oral: extensive first-pass metabolism (to oestrone); transdermal avoids first pass.
- Metabolised in liver (CYP3A4); enterohepatic circulation.
- Excreted in urine. **t½:** 13–20 h (oral).

PHARMACODYNAMICS

- Binds **oestrogen receptors** → relieves hot flushes, maintains bone density, restores vaginal epithelium.
- Unopposed oestrogen stimulates endometrial proliferation → hyperplasia/cancer.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Breast tenderness, nausea, headache.
- Fluid retention, bloating.
- Breakthrough bleeding.

ADVERSE EFFECTS (SERIOUS)

- VTE and stroke (oral > transdermal).
- Breast cancer (combined HRT, duration-related); endometrial cancer if unopposed.
- Gallbladder disease.
- Ovarian cancer (small ↑).

CONTRAINDICATIONS & CAUTIONS

- Current, past or suspected breast cancer; oestrogen-dependent tumours.
- Undiagnosed vaginal bleeding; untreated endometrial hyperplasia.
- Active or previous VTE; thrombophilia; recent arterial thrombosis (MI, stroke).
- Active liver disease; pregnancy.

KEY INTERACTIONS

- **Enzyme inducers:** ↓ effect.
- **Levothyroxine:** may ↑ requirements (oral).
- **Lamotrigine:** ↓ levels.

NURSING MANAGEMENT

- Review benefits/risks; check BP, BMI, breast screening attendance.
- Make sure progestogen prescribed if uterus present.
- Teach patch/gel application (clean, dry skin; rotate sites).
- Annual review.

PATIENT EDUCATION

- If you have a womb, take the progestogen part too.
- Report unexpected bleeding, breast lumps, leg swelling, chest pain.
- Attend breast screening.
- Patches: avoid breasts; rotate sites.

KEY POINTS

- Natural oestrogen for HRT.
- Transdermal has lower VTE risk.
- Progestogen needed if uterus intact.
- Breast cancer, VTE, endometrial cancer risks.

SOURCES CHECKED **[687]** Sandrena 1.0 mg gel (UK SmPC) **[688]** Estradiol 10 mcg Vaginal Tablets (UK SmPC) **[689]** Estradiol, oral (US label) *Full references p. 432–457*